

PAPER_691: 3D N-Body Gravitational Simulation: Euler and Leapfrog Integrators

Class: NBodySimulation3D

CP4 Entry: #275

Keywords: N-body simulation, Euler integration, leapfrog, energy conservation, gravitational softening

Session: 174 | **Version:** v5.31

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Abstract

A general-purpose 3D N-body gravitational simulation framework implementing both Euler and leapfrog (KDK) integration schemes. Includes softening parameter for numerical stability.

1. Equations of Motion

$$\mathbf{a}_i = \sum_{j \neq i} \frac{Gm_j(\mathbf{r}_j - \mathbf{r}_i)}{(|\mathbf{r}_{ij}|^2 + \epsilon^2)^{3/2}}$$

2. Euler Integrator

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\Delta t$$

$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \mathbf{a}(t)\Delta t$$

First-order accurate, $O(\Delta t)$. Energy drift proportional to Δt .

3. Leapfrog (KDK) Integrator

Half-kick: $\mathbf{v}_{n+1/2} = \mathbf{v}_n + \frac{1}{2}\mathbf{a}_n\Delta t$ Drift: $\mathbf{r}_{n+1} = \mathbf{r}_n + \mathbf{v}_{n+1/2}\Delta t$ Kick: $\mathbf{v}_{n+1} = \mathbf{v}_{n+1/2} + \frac{1}{2}\mathbf{a}_{n+1}\Delta t$ Second-order symplectic — conserves phase-space volume exactly.

4. Total Energy Conservation

$$E_{total} = \sum_i \frac{1}{2}m_i v_i^2 - \sum_{i < j} \frac{Gm_i m_j}{r_{ij}}$$

5. Softening

Gravitational softening ϵ prevents divergences at small separations. Typical: $\epsilon \sim 1$ kpc for galaxy-scale simulations.

References

- Press et al. (1992), Numerical Recipes
- UQFF Framework, Star-Magic Session 174